

A. Rollout materialization

State and table

$s_t, \mathcal{Q}_t$
masks and reasons

Oracle score

target RRI for
every valid row

Select and step

choose valid a_t
fuse selected geometry
regenerate $\mathcal{Q}_{t+1}$

Replay row

row labels plus
selected lineage

B. Masked Double-Q target

Replay sample

$s, \mathcal{Q}, m$
 $s', \mathcal{Q}', m'$

Online selector

$i^* = \arg \max_{j: m'j=1}$
 $Q_{\theta(s', q'_j)}$

Target backup

$Q_{|(\theta)}(s', q'_{i^*})$
 $y = r^e + \gamma Q_{|(\theta)}(\cdot)$
masked loss

Held-out policy

deploy π_Q
oracle-rescore trajectory

All valid rows can receive one-step labels; only the selected row creates successor observations.